

MATTHEW YIZHENG HE

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Education

McGill University, Montreal, Canada

2023 – 2027 (expected)

B.A. & Sc. Honours Cognitive Science (Computer Science Focus), Major Mathematics

CGPA: 4.0/4.0 (Top 5%)

Research Experience

Masset Lab *Research Intern*

↔ Prof. Paul Masset

2024–Present

McGill University, Mila–Quebec AI Institute

Projects involved:

- **Simultaneous detection and estimation in olfactory sensing (Sep. 2024–Present)**
(Recurrent neural networks; Constrained optimization; Bayesian inference; Mirror-Langevin descent)
Designed, coded, and ran scaling experiments on HPC clusters; refined model variants to improve performance.
First author on the accepted abstract and co-first author on the associated preprint (under review at PNAS).
- **Population structure of reward-induced remapping in hippocampal CA1 (Apr. 2025–Present)**
(Representation similarity analysis (RSA); Tensor decomposition; Dimensionality reduction techniques)
Built data pipelines for preprocessed data; performed exploratory data analysis and developed metrics to quantify neural population structure and reward-related remapping. Made figures for the accepted COSYNE abstract.
- **Piecewise deterministic spiking samplers (Nov. 2025–Present)**
(Piecewise-deterministic Markov processes; MCMC sampling algorithms; Spiking neural networks)
Participating in both simulation-based and theoretical investigation.

Laboratory for Neural Computation and Adaptation *Research Intern*

↔ Prof. Taro Toyozumi

June, 2026–August 2026

RIKEN Center for Brain Science

Research Interests

Machine Learning Theory (High dimensional learning dynamics); Theoretical & Computational Neuroscience

Preprint

Chen Jiang*, **Matthew Y. He***, Venkatesh N. Murthy, Cengiz Pehlevan, Jacob A. Zavatone-Veth[†], and Paul Masset[†] (2025). “Simultaneous detection and estimation in olfactory sensing”. In: *bioRxiv*. (Under review at PNAS). doi: [10.1101/2025.11.01.686013](https://doi.org/10.1101/2025.11.01.686013).

* Equal contribution. [†] Co-supervision.

Conference Abstract

Matthew He, Chen Jiang, Venkatesh Murthy, Cengiz Pehlevan, Jacob Zavatone-Veth, and Paul Masset (2025). “Simultaneous detection and mapping in the olfactory bulb”. In: *Computational and Systems Neuroscience (COSYNE)*.

Chen Jiang, Adel Halawa, **Matthew He**, and Paul Masset (2026). “Population structure of reward-induced remapping in the hippocampal CA1”. In: *Computational and Systems Neuroscience (COSYNE)*.

Awards & Distinctions

Faculty of Science Scholarship

2025-2026

Established in 1992 by the University to provide awards based on academic achievement to students in the top 5% of the Faculty of Science.

Dean’s Honour List

2023-2025

Top 10% of students in the Faculty of Science based on the combined GPA for the fall and winter terms.

Science Undergraduate Research Awards

2025

\$8950 CAD for 15 weeks of full-time research in the summer of 2025.

SUS Ambassador Fund

2025

\$750 CAD for traveling to present at Computational and Systems Neuroscience (COSYNE) 2025 conference.

Tomlinson Undergraduate Award

2025

Mentoring students in a molecular and cellular biology course.

Conference & Workshop Attendance

Mathematical Foundation for Data Science *Stochastic Optimization; High-dimensional Statistics* **May–June 2025**
↪ Centre de Recherches Mathématiques, Montreal, Canada

Computational and Systems Neuroscience (COSYNE) 2025 *Conference & Workshop* **April 2025**

SynBio Collective Wet Lab Workshop Series *Theme: Expressing recombinant genes in bacteria* **January–April 2024**
↪ McGill iGEM Team

Selected Coursework

Graduate-level	Undergraduate-level (selected)
Theory of Machine Learning (MATH562)	Nonlinear Optimization (MATH378)
Representation Learning (IFT6135)	Honours Algorithms & Data Structures (COMP252)
Honours Convex Optimization (MATH563)	Honours Analysis 2 (MATH255)
Stochastic Processes (MATH547)	Honours Nonlinear Dynamics (MATH376)
Numerical Analysis (MATH578)	Probability & Statistics (MATH323,324)
Computational Neuroscience (NEUR503)	Honours Applied Linear Algebra (MATH247)
	Discrete Structures (MATH240)
	Cognition (PSYC213)

Skills

Scientific Computing	Python (PyTorch, JAX), MATLAB, Linux, HPC clusters (SLURM)
Mathematical Areas (beyond coursework)	Bayesian inference, sampling methods, tensor decomposition, physics-informed neural networks (PINNs)
Neuroscience Knowledge	Hippocampus, olfactory bulb, midbrain dopaminergic neurons; general systems neuroscience knowledge learned through courses